MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

RELAY, SIGNAL, KEY CODE HRQ

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic signal relay module.

1.2 Classification. This specification covers two classes of modules.

Class I and class II as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

* 2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

MIL-R-39016 - Relays, Electromagnetic, Established Reliability.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.15 failures per million operating hours for class I or 0.27 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, failure rate for class I shall be used for reliability prediction (see 6.3).

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I. Class I modules shall conform to the electrical requirements of class I only, and class II modules shall conform to the electrical requirements of class I and class II (see 6.3).

3.4.1 Relay qualification requirements. All relays shall conform to the requirements of MIL-R-39016.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be HRQ.

3.5.2 Size. The module size shall be 1B.

3.5.3 Weight. The module shall have a maximum weight of 70 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction fin on increment A and heat dissipation fins on increments A and B.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/311-1 for class I and M28787/311-2 for class II, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

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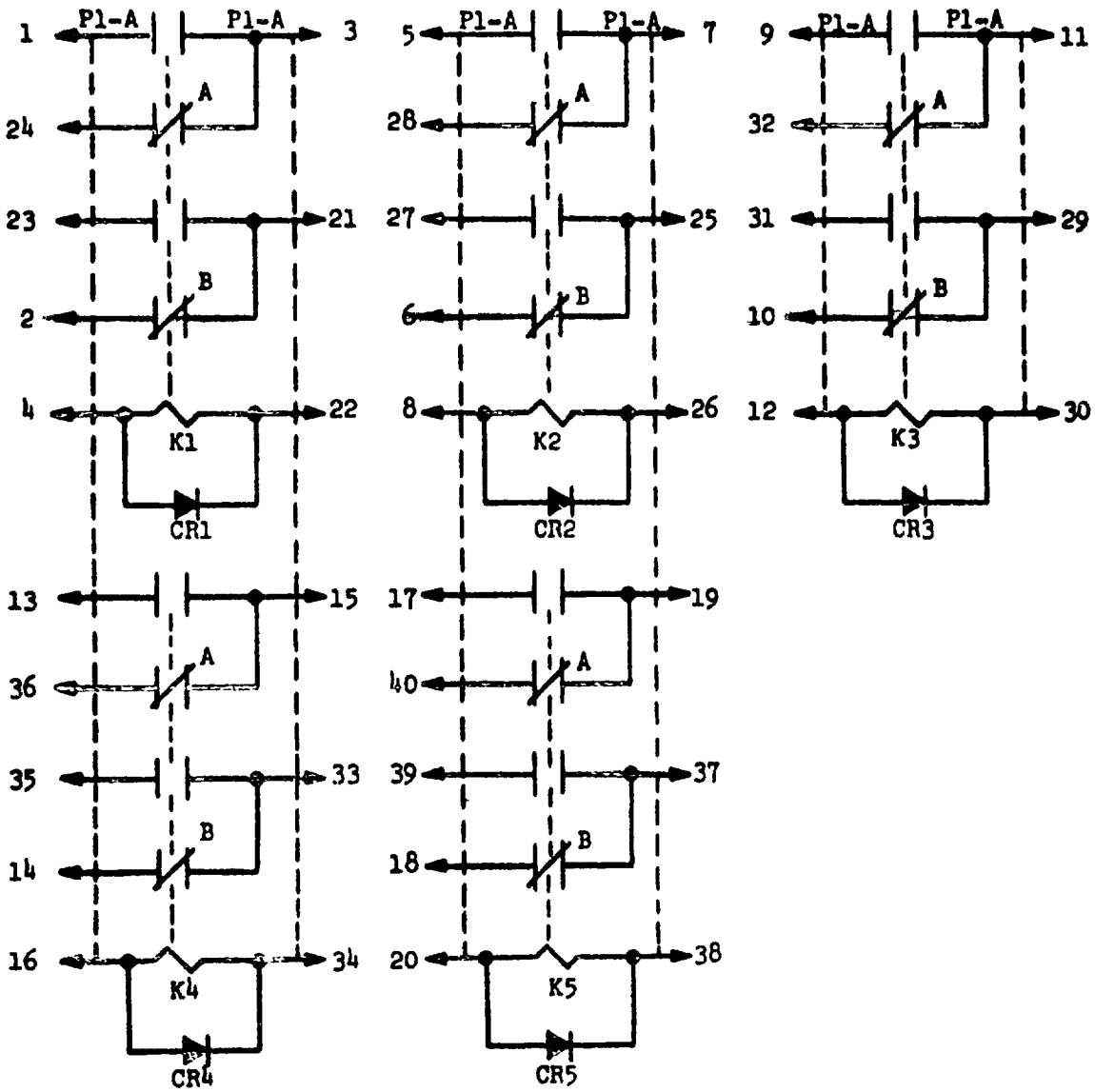


FIGURE 1. HRQ module function diagram, relay.

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* TABLE I. Module electrical requirements.

Test conditions	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		85° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A1-A22	Isolation													
	Initial	II	S	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	II	S	1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
B1-B20	Open contact resistance													
	Initial	III	X	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	III	X	1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
C1-C20	Closed contact resistance check													
	Initial	IV	X	0.00	0.16	0.00	0.16	0.00	0.16	0.00	0.16	0.00	0.16	Ω
	End-of-life	IV	X	0.0	0.5	0.0	0.5	0.0	0.5	0.0	0.5	0.0	0.5	Ω
D1-D5	Diode forward voltage drop	V	X	100	840	100	890	100	770	100	1000	100	690	mV
E1-E5	Coil resistance check	VI	X	765	935	682	852	882	1118	498	668	1014	1184	Ω
F1-F10	Pickup voltage													
	Initial	VII	X	4	7	4	8	4	8	4.0	8.5	4.0	8.5	V
	End-of-life	VII	X	4.0	7.5	4.0	8.5	4.0	8.5	4	9	4	9	V
G1-G10	Dropout voltage													
	Initial	VIII	X	0.64	4.00	0.64	4.00	0.64	4.00	0.64	4.00	0.64	4.00	V
	End-of-life	VIII	X	0.41	6.50	0.41	6.50	0.41	6.50	0.41	6.50	0.41	6.50	V
H1-H10	Operate time	IX	Q	1	3	1	4	1	4	1	4	1	4	ms
J1-J10	Release time	X	Q	1	6	1.0	7.5	1.0	7.5	1.0	7.5	1.0	7.5	ms
K1-K20	Contact bounce	XI	Q	0.0	1.5	0.0	1.5	0.0	1.5	0.0	1.5	0.0	1.5	ms

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.
4/ Unless otherwise specified, limits are for initial and end-of-life.

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4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through K shall be performed in any order with the following exceptions. If test condition A is part of a sequence of tests, the order of testing shall be as follows: A, where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through K in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B through K shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall not be operated. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition A, shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Coil life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the coil life test and while still at the maximum class temperature, electrical tests B through K shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests B through K shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I. The module shall then be subjected to the overload test of 4.5.5.

4.5.4 Contact life test. Each relay shall be subjected to a minimum of 100,000 cycles operated as shown on figure 3 at the maximum class temperature. Following the contact life test, test conditions B through K shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I. A module failure is defined as a failure to meet any of the end-of-life limits of table I. Contact current shall not exceed 10 mA during post contact life testing. Coil life and contact life shall not be run on the same module.

4.5.5 Overload test. The relay shall be subjected to 100 cycles at 10 Hz as shown on figure 4 at the maximum class temperature. Following the return to 25°C ambient temperature, test conditions B through K shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

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6.3 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in a class I environment, the failure rate for class I shall be used for reliability prediction (see 3.3 and 3.4).

* 6.4 Module description. The following electrical parameters reflect the 0° to 60°C and -55° to 85°C end-of-life limits from table I.

	Class I	Class II	Units
Maximum power dissipation	2.5	2.5	W
Coil voltage and current per relay	20.0 V at 30 mA (max)	20.0 V at 40 mA(max)	
Contact rating	28 V at 1 A (max)	28 V at 1 A (max)	
Pickup voltage	4.0 to 8.5	4.0 to 9.0	V
Dropout voltage	0.64 to 4.0	0.64 to 6.5	V
Operate time	1.0 to 4.0	1.0 to 4.0	ms
Release time	1.0 to 7.5	1.0 to 7.5	ms

* 6.5 Module function. This module contains five relays, each with two normally-open and two normally-closed contacts. Across each relay coil is a diode for suppression of switching transients.

* 6.5.1 Module operating constraints. When operating one, two, or three relays per module simultaneously, the maximum voltage which may be applied to the coil is 20 V per relay. When operating four or five relays per module simultaneously, the maximum coil voltage must be reduced to stay within the module maximum power dissipation of 2.5 W.

6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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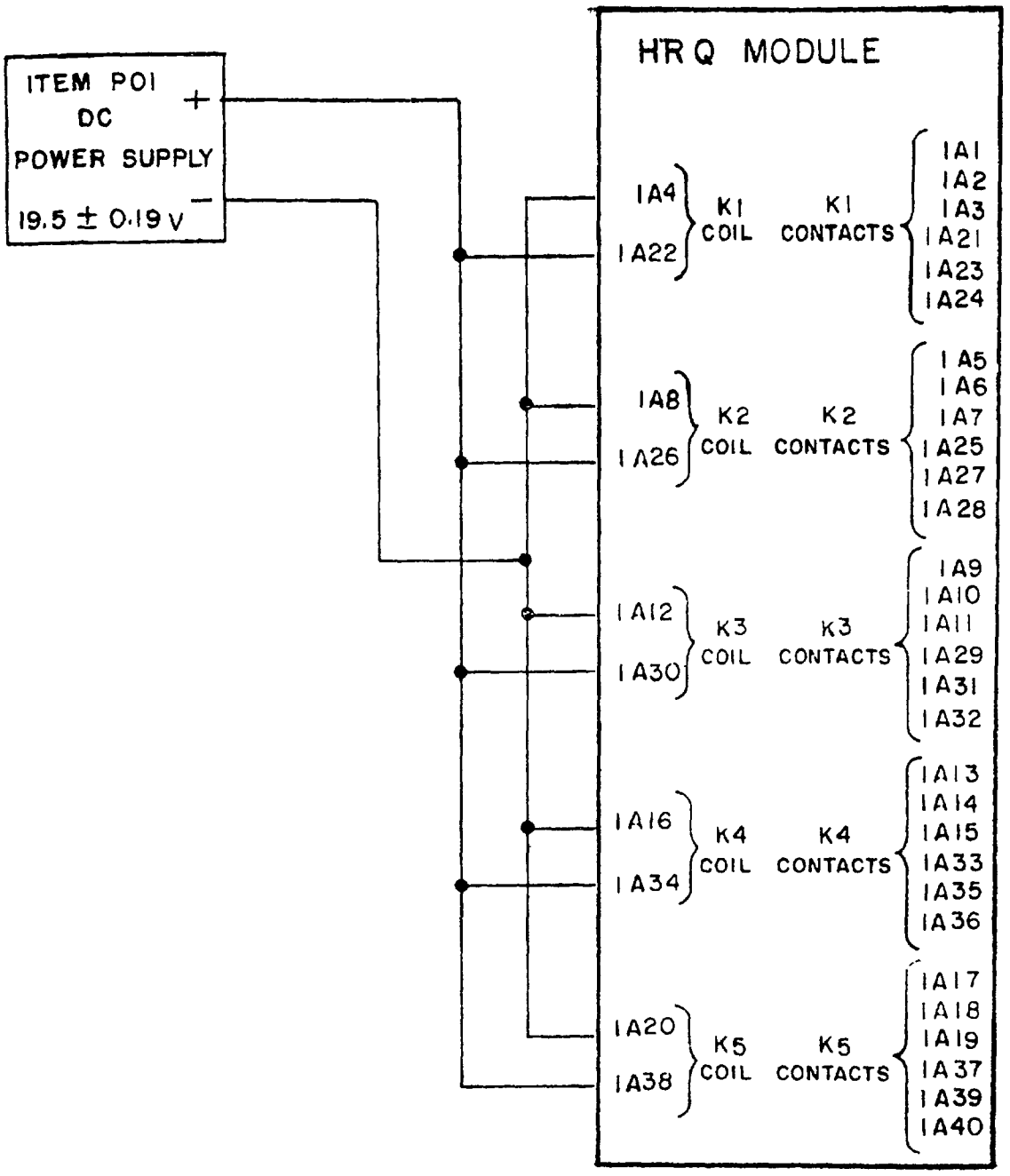
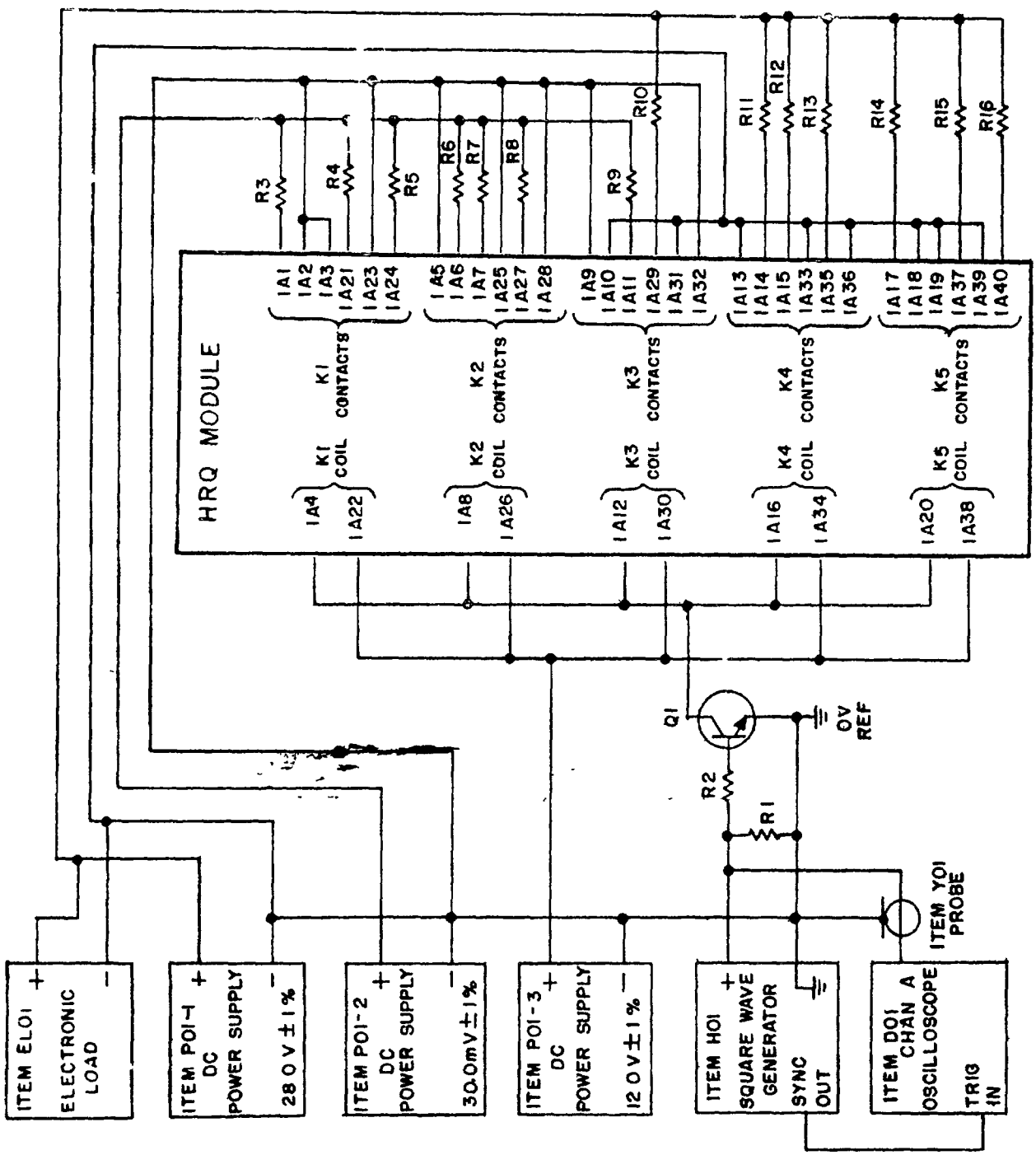


FIGURE 2. Operating temperature, component temperature and coil life test circuit.

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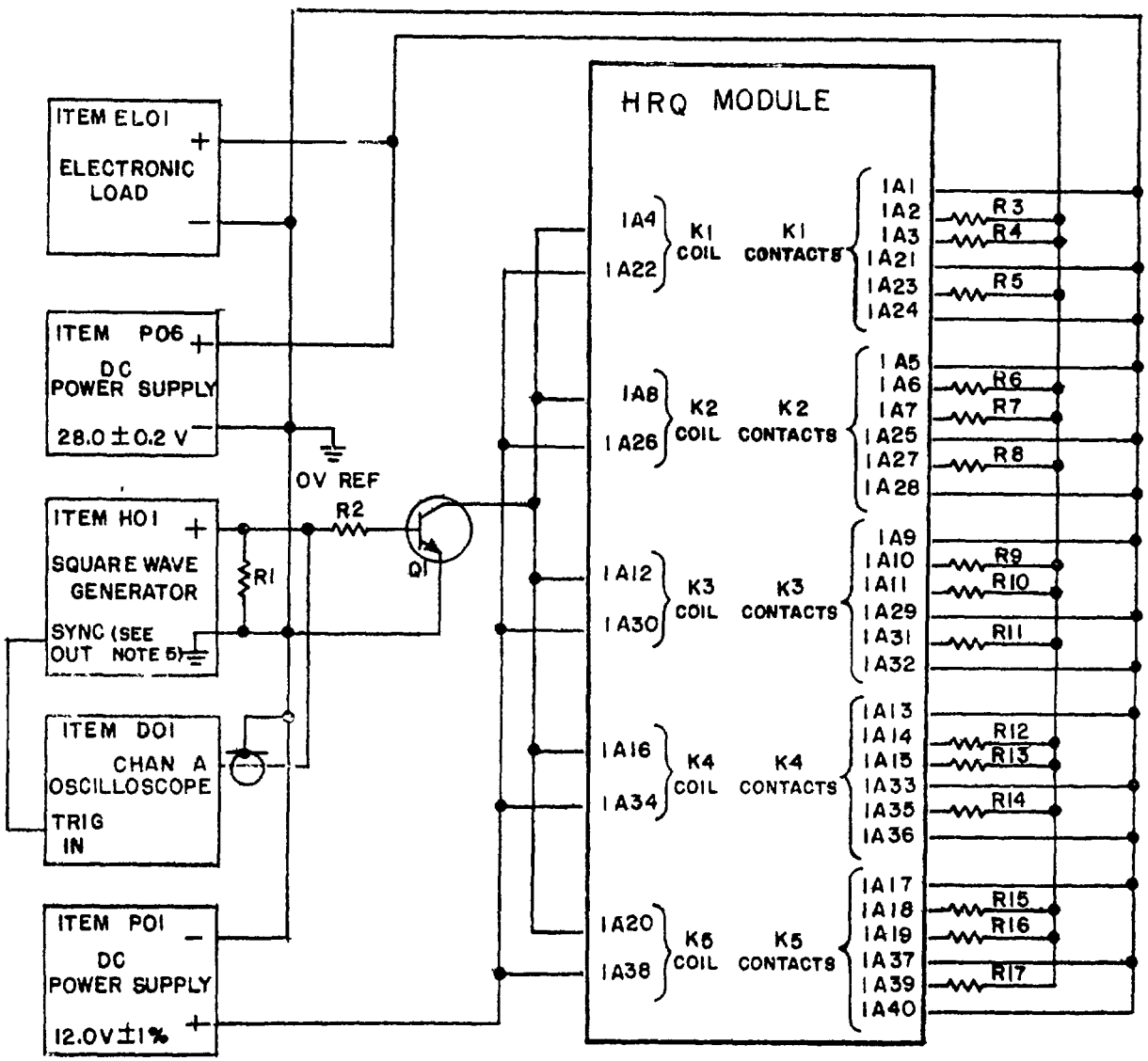
* FIGURE 3. Contact life test circuit.

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NOTES:
1. R1 = 49.9Ω ±5%, 1 W.
2. R2 = 499Ω ±1%, 1/10 W.
3. R3-R9 = 28.0Ω ±1%, 30 W.
4. R10-R16 = 3.0Ω ±1%, 1/10W.
5. Q1 = 2N3108 or equivalent.
6. Item H01 setup:
   PA = 2.5 V peak as measured on item D01.
   Frequency = 10 Hz ±1%.
7. Adjust item EL01 for a current of 100 mA

FIGURE 3. Contact life test circuit - Continued.

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NOTES:
1. R1 = 49.9Ω ±5%, 1 W.
2. R2 = 499Ω ±1%, 1/10 W.
3. R3-R17 = 14.0Ω ±5%, 70 W
4. Q1 = 2N3108 or equivalent
5. Item H01 setup:
   PA = 2.5 V peak as read on item D01.
   Frequency = 10.0 Hz ±1%.
6. Adjust item EL01 for a current of 2.0A ±10%.

FIGURE 4. Overload test circuit.

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* TABLE II. Isolation test.

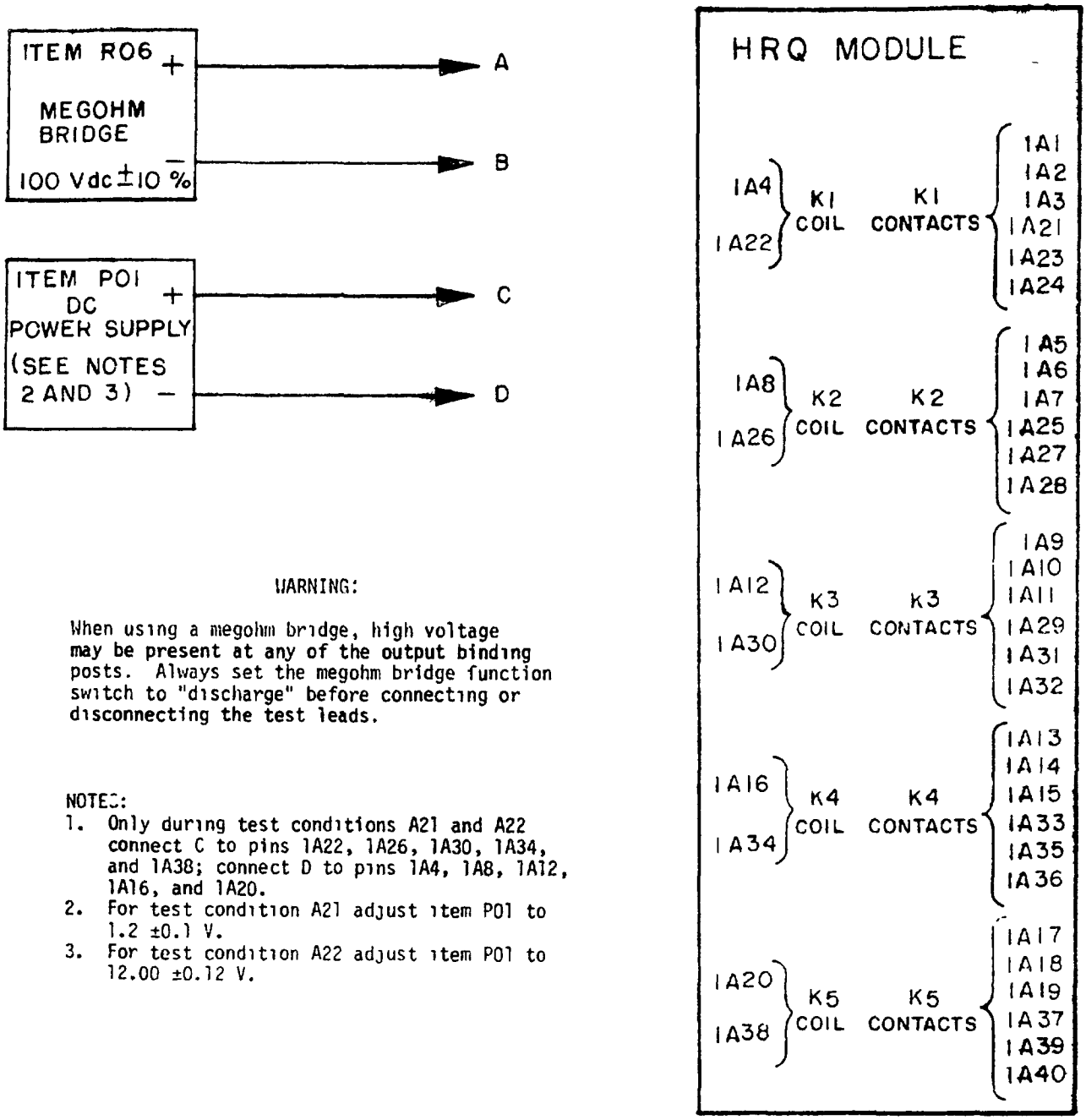
Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1	Isolation	1A1,1A3,1A21, 1A23	1A4	10	∞	MΩ	5
A2		1A13,1A15, 1A33,1A35	1A16				
A3		1A5,1A7,1A25, 1A27	1A8				
A4		1A17,1A19, 1A37,1A39	1A20				
A5		1A9,1A11,1A29, 1A31	1A12				
A6		1A1,1A3	1A21,1A23				
A7		1A13,1A15	1A33,1A35				
A8		1A5,1A7	1A25,1A27				
A9		1A17,1A19	1A37,1A39				
A10		1A9,1A11	1A29,1A31				
A11		1A1-1A4, 1A21-1A24	1A13-1A16, 1A33-1A36				
A12		1A1-1A4, 1A21-1A24	1A5-1A8, 1A25-1A28				
A13		1A1-1A4, 1A21-1A24	1A17-1A20, 1A37-1A40				
A14		1A1-1A4, 1A21-1A24	1A9-1A12, 1A29-1A32				
A15		1A13-1A16, 1A33-1A36	1A5-1A8, 1A25-1A28				
A16		1A13-1A16, 1A33-1A36	1A17-1A20, 1A37-1A40				
A17		1A13-1A16, 1A33-1A36	1A9-1A12, 1A29-1A32				
A18		1A5-1A8, 1A25-1A28	1A17-1A20, 1A37-1A40				

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* TABLE II. Isolation test - Continued.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A19	Isolation	1A5-1A8, 1A25-1A28	1A9-1A12, 1A29-1A32	10	∞	MΩ	5
A20		1A17-1A20, 1A37-1A40	1A9-1A12, 1A29-1A32				
A21		1A3,1A7,1A11, 1A15,1A19, 1A21,1A25, 1A29,1A33,1A37	1A1,1A5,1A9, 1A13,1A17, 1A23,1A27, 1A31,1A35,1A39				
A22		1A3,1A7,1A11, 1A15,1A19, 1A21,1A25, 1A29,1A33,1A37	1A2,1A6,1A10, 1A14,1A18, 1A24,1A28, 1A32,1A36,1A40				

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WARNING:

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTES:
1. Only during test conditions A21 and A22 connect C to pins 1A22, 1A26, 1A30, 1A34, and 1A38; connect D to pins 1A4, 1A8, 1A12, 1A16, and 1A20.
2. For test condition A21 adjust item PO1 to 1.2 ±0.1 V.
3. For test condition A22 adjust item PO1 to 12.00 ±0.12 V.

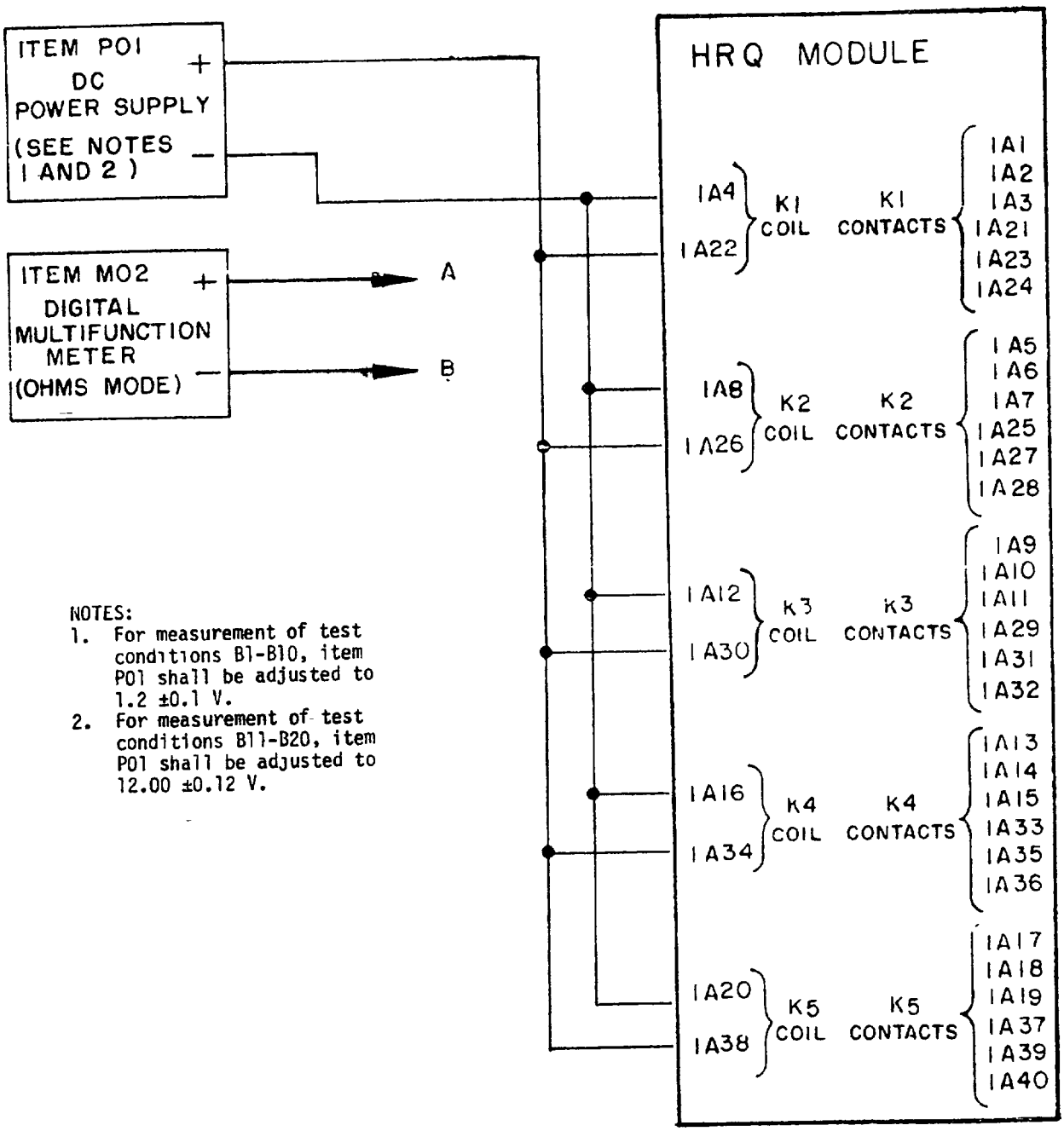
FIGURE 5. Isolation test circuit for table II.

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TABLE III. Open contact resistance test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B1	Open contact resistance	1A3	1A1	10	∞	MΩ	6
B2	↓	1A21	1A23	↓	↓	↓	↓
B3		1A7	1A5				
B4		1A25	1A27				
B5		1A11	1A9				
B6		1A29	1A31				
B7		1A15	1A13				
B8		1A33	1A35				
B9		1A19	1A17				
B10		1A37	1A39				
B11		1A3	1A24				
B12		1A21	1A2				
B13		1A7	1A28				
B14		1A25	1A6				
B15		1A11	1A32				
B16		1A29	1A10				
B17		1A15	1A36				
B18		1A33	1A14				
B19		1A19	1A40				
B20		1A37	1A18				

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NOTES:
1. For measurement of test conditions B1-B10, item P01 shall be adjusted to 1.2 ±0.1 V.
2. For measurement of test conditions B11-B20, item P01 shall be adjusted to 12.00 ±0.12 V.

FIGURE 6. Open contact resistance test circuit for table III.

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* TABLE IV. Closed contact resistance check test.

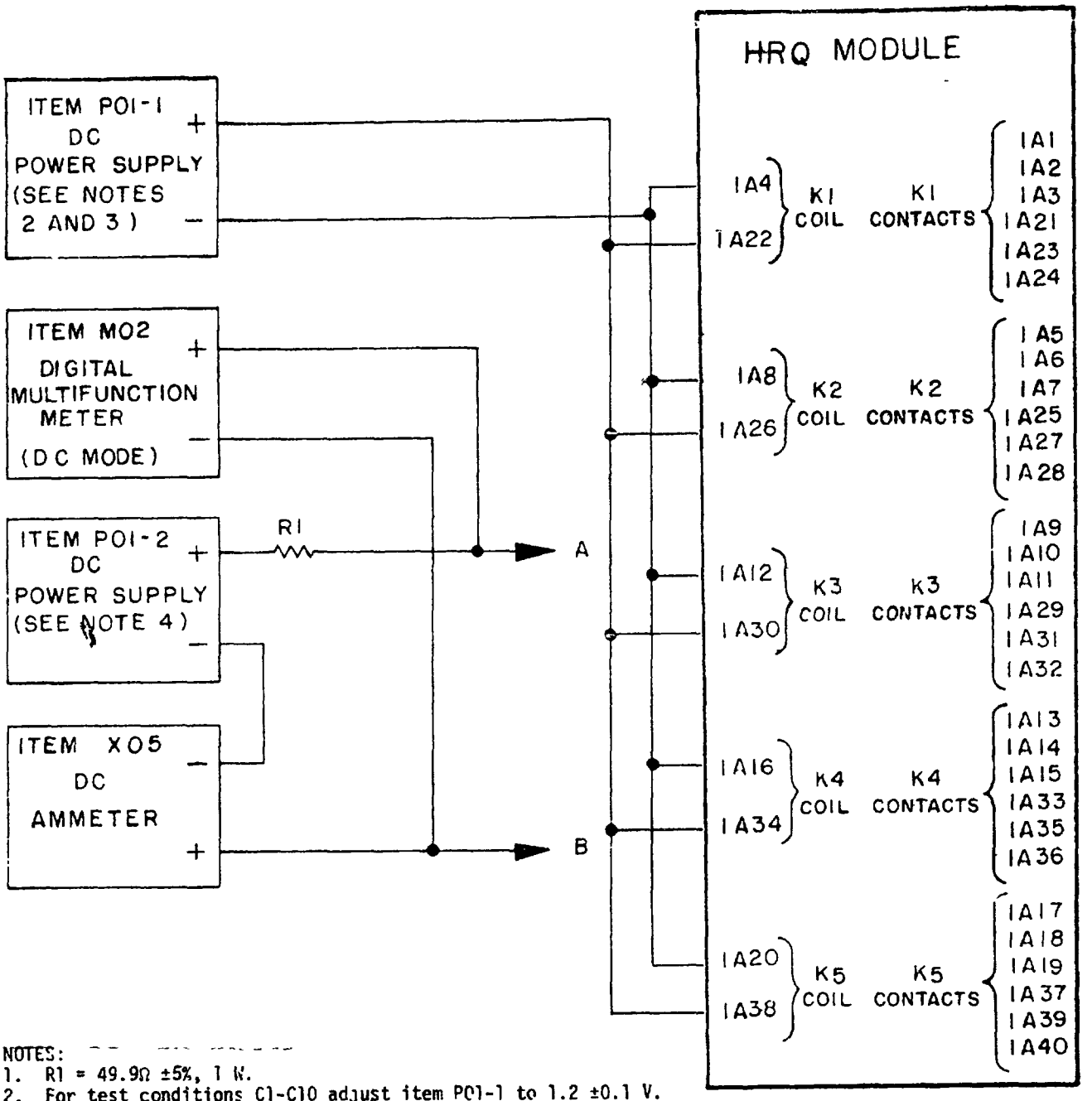
Test condition	Parameter	Connect		Initial limits 25° ± 5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C1	Closed contact resistance check 1/	1A3	1A24	0	16	mV	7
C2	↓	1A21	1A2	↓	↓	↓	↓
C3		1A15	1A36				
C4		1A33	1A14				
C5		1A7	1A28				
C6		1A25	1A6				
C7		1A19	1A40				
C8		1A37	1A18				
C9		1A11	1A32				
C10		1A29	1A10				
C11		1A3	1A1				
C12		1A21	1A23				
C13		1A15	1A13				
C14		1A33	1A35				
C15		1A7	1A5				
C16		1A25	1A27				
C17		1A19	1A17				
C18		1A37	1A39				
C19		1A11	1A9				
C20	↓	1A29	1A31	↓	↓	↓	↓

1/ The closed contact resistance in ohms can be determined from the following equation:

$$R = \frac{V}{I} = \frac{V}{100}$$

where, V = the measured voltage across the relay contact in mV,
and I = the current through the relay.

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NOTES:
1. R1 = 49.9Ω ±5%, 1 W.
2. For test conditions C1-C10 adjust item P01-1 to 1.2 ±0.1 V.
3. For test conditions C11-C20 adjust item P01-1 to 12.0 ±0.12 V.
4. For test conditions C1-C20 adjust item P01-2 for a current of 100 mA ±1% as measured on item X05 (see note 6).
5. Voltage sensing leads from the positive and negative terminals of item M02 to the module pins at points A and B shall be connected in such a way as to exclude the resistance (or voltage drop at 100 mA) of the current-carrying leads.
6. For post contact life test, item P01-2 shall be adjusted for a current of 10 $^{+0.0}_{-0.1}$ mA as measured on item X05. P01-2 shall not exceed 0.55 V. The limits of table I shall be converted to item M02 voltage reading by the note in table IV.

* FIGURE 7. Closed contact resistance check test circuit for table IV.

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* TABLE V. Diode forward voltage drop test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
D1	Diode forward voltage drop	1A4	Open	1A22	A	100	840	mV	8
D2	↓	1A16	↓	1A34	↓	↓	↓	↓	↓
D3		1A8		1A26					
D4		1A20		1A38					
D5		1A12		1A30					

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* TABLE VI. Coil resistance check test.

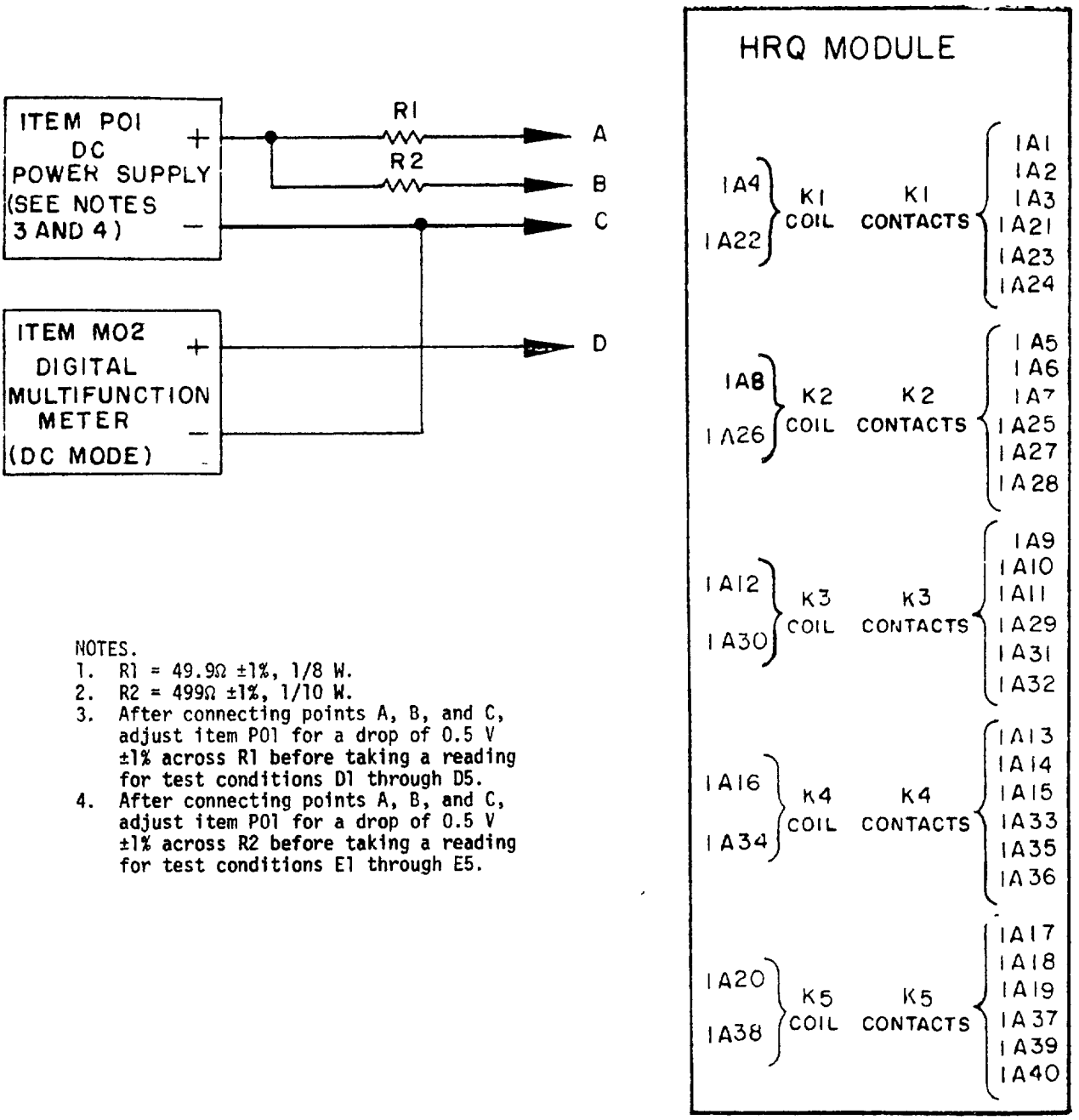
Test condition	Parameter	Connect				Initial limits 25° ± 5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
E1	Coil resistance check 1/	Open	1A22	1A4	B	765	935	mV	8
E2	↓	↓	1A34	1A16	↓	↓	↓	↓	↓
E3			1A26	1A8					
E4			1A38	1A20					
E5			1A30	1A12					

1/ The coil resistance in ohms can be determined from the following equation:

$$R = \frac{V}{I} = \frac{mV}{1.0\ mA}$$

where, mV = the measured voltage across the relay coil.

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NOTES.
1. R1 = 49.9Ω ±1%, 1/8 W.
2. R2 = 499Ω ±1%, 1/10 W.
3. After connecting points A, B, and C, adjust item P01 for a drop of 0.5 V ±1% across R1 before taking a reading for test conditions D1 through D5.
4. After connecting points A, B, and C, adjust item P01 for a drop of 0.5 V ±1% across R2 before taking a reading for test conditions E1 through E5.

* FIGURE 8. Diode forward voltage and coil resistance check test circuit for tables V and VI.

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TABLE VII. Pickup voltage test.

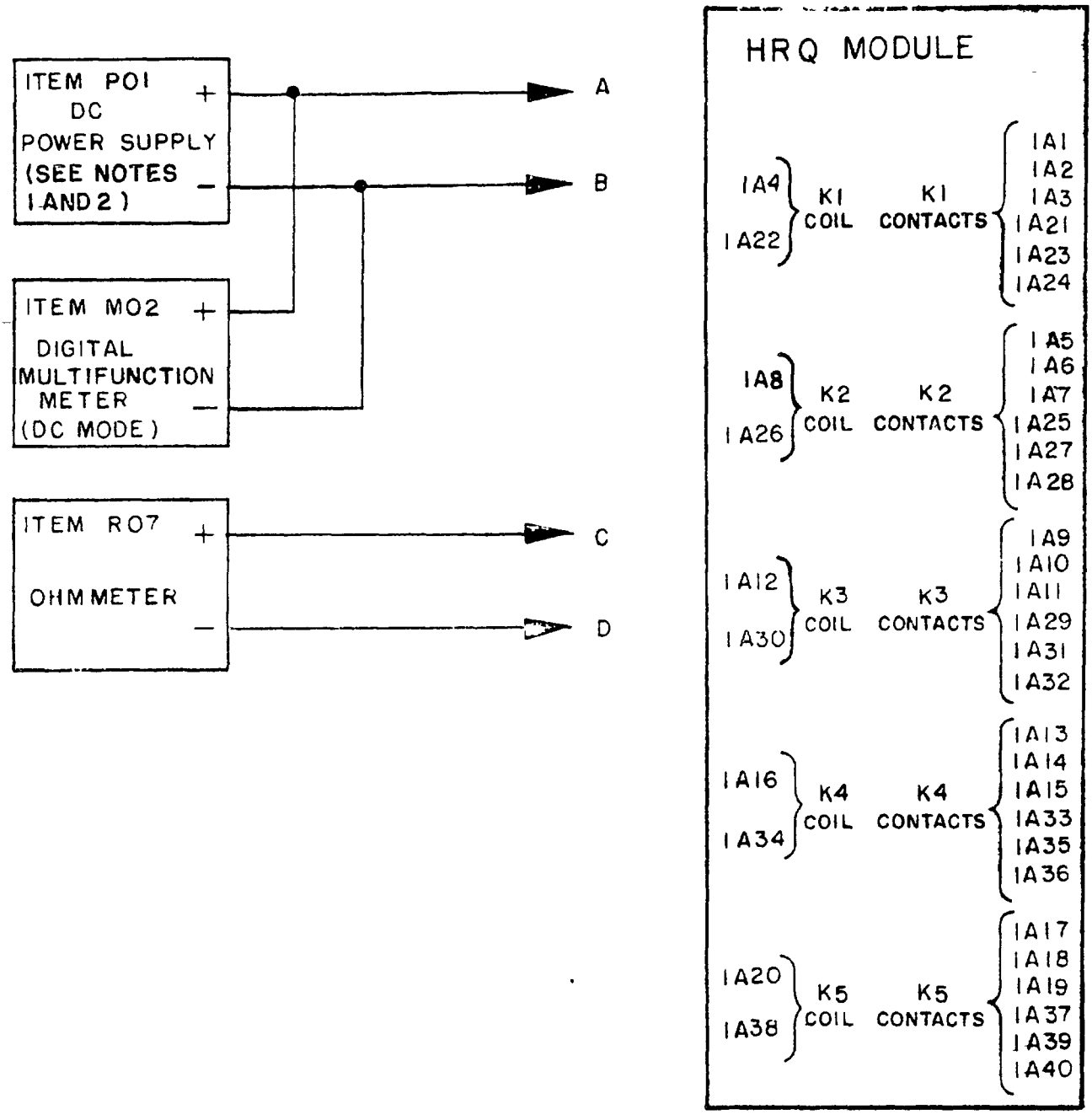
Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
F1	Pickup voltage	1A22	1A4	1A3	1A1	4	7	V	9
F2	↓	1A22	1A4	1A21	1A23	↓	↓	↓	↓
F3		1A34	1A16	1A15	1A13				
F4		1A34	1A16	1A33	1A35				
F5		1A26	1A8	1A7	1A5				
F6		1A26	1A8	1A25	1A27				
F7		1A38	1A20	1A19	1A17				
F8		1A38	1A20	1A37	1A39				
F9		1A30	1A12	1A11	1A9				
F10		1A30	1A12	1A29	1A31				

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TABLE VIII. Dropout voltage test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
G1	Dropout voltage	1A22	1A4	1A3	1A24	0.64	4.00	V	9
G2	↓	1A22	1A4	1A21	1A2	↓	↓	↓	↓
G3		1A34	1A16	1A15	1A36				
G4		1A34	1A16	1A33	1A14				
G5		1A26	1A8	1A7	1A28				
G6		1A26	1A8	1A25	1A6				
G7		1A38	1A20	1A19	1A40				
G8		1A38	1A20	1A37	1A18				
G9		1A30	1A12	1A11	1A32				
G10		1A30	1A12	1A29	1A10				

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* FIGURE 9. Pickup voltage and dropout voltage test circuit for tables VII and VIII.

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NOTES:
1. Test procedure for test conditions F1-F10:

   a. Connect test points A, B, C, and D as specified for test condition F1.
   b. Adjust output of item P01 for 19.5 ±0.5 V as measured on item M02. Maintain this voltage for a period of 1 to 3 seconds.
   c. Gradually reduce the output voltage of item P01 to zero. The contact resistance, as measured on item R07, should be 1 megohm or greater.
   d. Gradually increase the output voltage of item P01 (do not exceed 20 V ±0%) until the contact resistance, as measured on item R07, drops to 1 ohm or less.
   e. The pickup voltage shall be read on item M02 and shall be as specified for test condition F1.
   f. Repeat notes a through e for test conditions F2-F10.

2. Test procedure for test conditions G1-G10:

   a. Connect test points A, B, C, and D as specified for test condition G1.
   b. Adjust output of item P01 for 19.5 ±0.5 V as measured on item M02. The contact resistance, as measured on item R07, should be 1 megohm or greater.
   c. Gradually reduce the output voltage of item P01 until the contact resistance, as measured on item R07, drops to 1 ohm or less.
   d. The dropout voltage shall be read on item M02 and shall be as specified for test condition G1.

e. Repeat notes a through d for test conditions G2-G10.

FIGURE 9. Pickup voltage and dropout voltage test circuit for tables VII and VIII - Continued.

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TABLE IX. Operate time test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
H1	Operate time	1A3	1A1	1	3	ms	10, 11
H2		1A21	1A23				
H3		1A15	1A13				
H4		1A33	1A35				
H5		1A7	1A5				
H6		1A25	1A27				
H7		1A19	1A17				
H8		1A37	1A39				
H9		1A11	1A9				
H10		1A29	1A31				

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TABLE X. Release time test.

Test condition	Parameter	Connect		Initial limits 25° ± 5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
J1	Release time	1A3	1A24	1	6	ms	10, 11
J2		1A21	1A2				
J3		1A15	1A36				
J4		1A33	1A14				
J5		1A7	1A28				
J6		1A25	1A6				
J7		1A19	1A40				
J8		1A37	1A18				
J9		1A11	1A32				
J10		1A29	1A10				

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TABLE XI. Contact bounce test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
K1	Contact bounce	1A3	1A1	0.0	1.5	ms	10, 11
K2		1A21	1A23				
K3		1A15	1A13				
K4		1A33	1A35				
K5		1A7	1A5				
K6		1A25	1A27				
K7		1A19	1A17				
K8		1A37	1A39				
K9		1A11	1A9				
K10		1A29	1A31				
K11		1A3	1A24				
K12		1A21	1A2				
K13		1A15	1A36				
K14		1A33	1A14				
K15		1A7	1A28				
K16		1A25	1A6				
K17		1A19	1A40				
K18		1A37	1A18				
K19		1A11	1A32				
K20		1A29	1A10				

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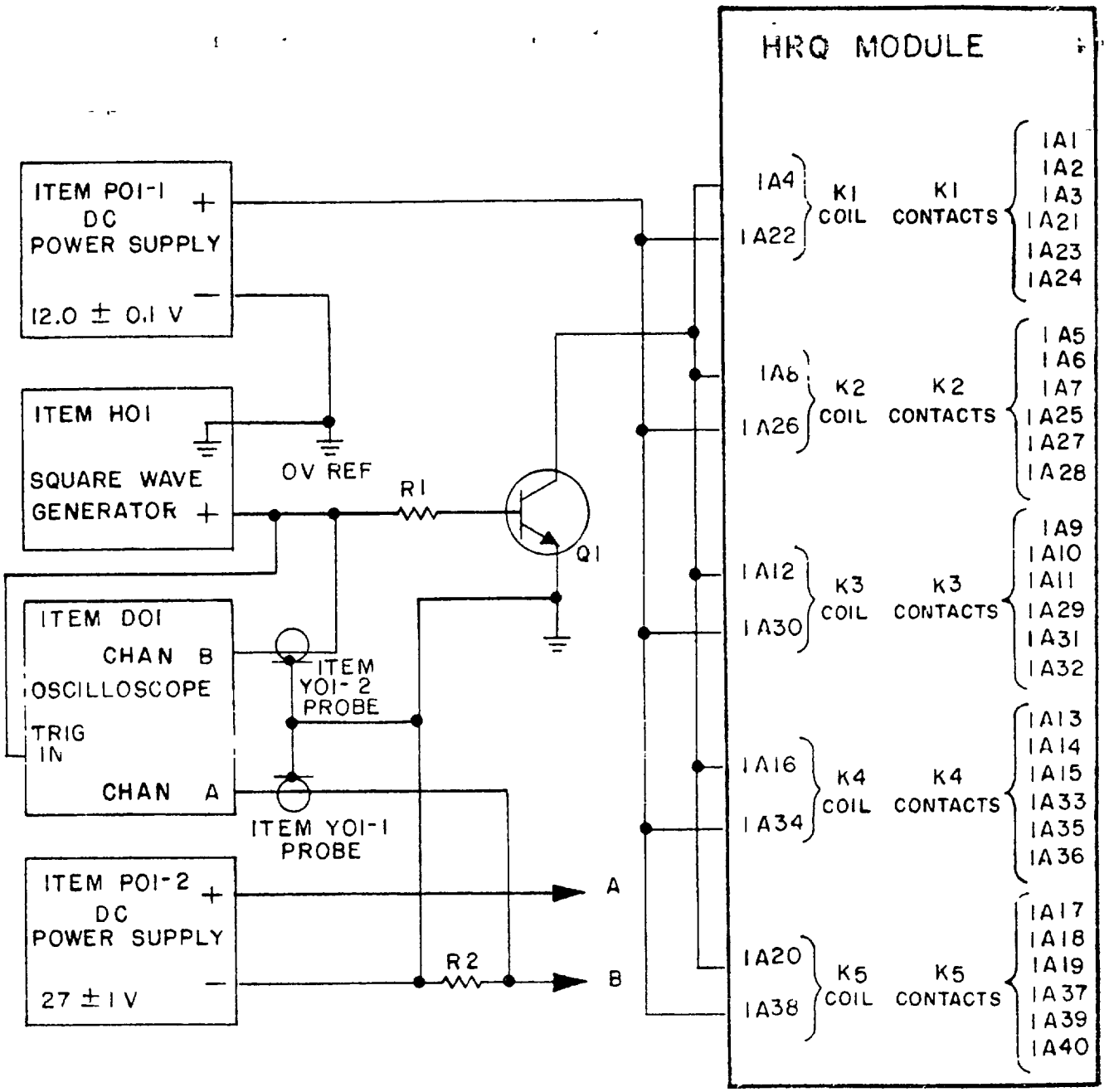


FIGURE 10. Operate time, release time, and contact bounce test circuit for tables IX, X, and XI.

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NOTES:
1. R1 = 499Ω ±1%, 1/10 W.
2. R2 = 28 kΩ ±1%, 1/4 W.
3. Q1 = 2N3108, or equivalent.
4. Test procedure for test conditions H1-H10.

   a. Connect test points A and B as specified for test condition H1.
   b. On item D01, set the trigger coupling to dc, the trigger source to external, and the trigger level and slope to posttive.
   c. Adjust output of item H01 for a frequency of 9.0 Hz ±10% and an amplitude of 2.5 ±0.1 Vp as read on item D01, channel B.
   d. The operate time shall be read on item D01, channel A, and shall be as specified for test condition H1.
   e. Repeat notes a through d for test conditions H2-H10.

5. Test procedure for test conditions J1-J10:

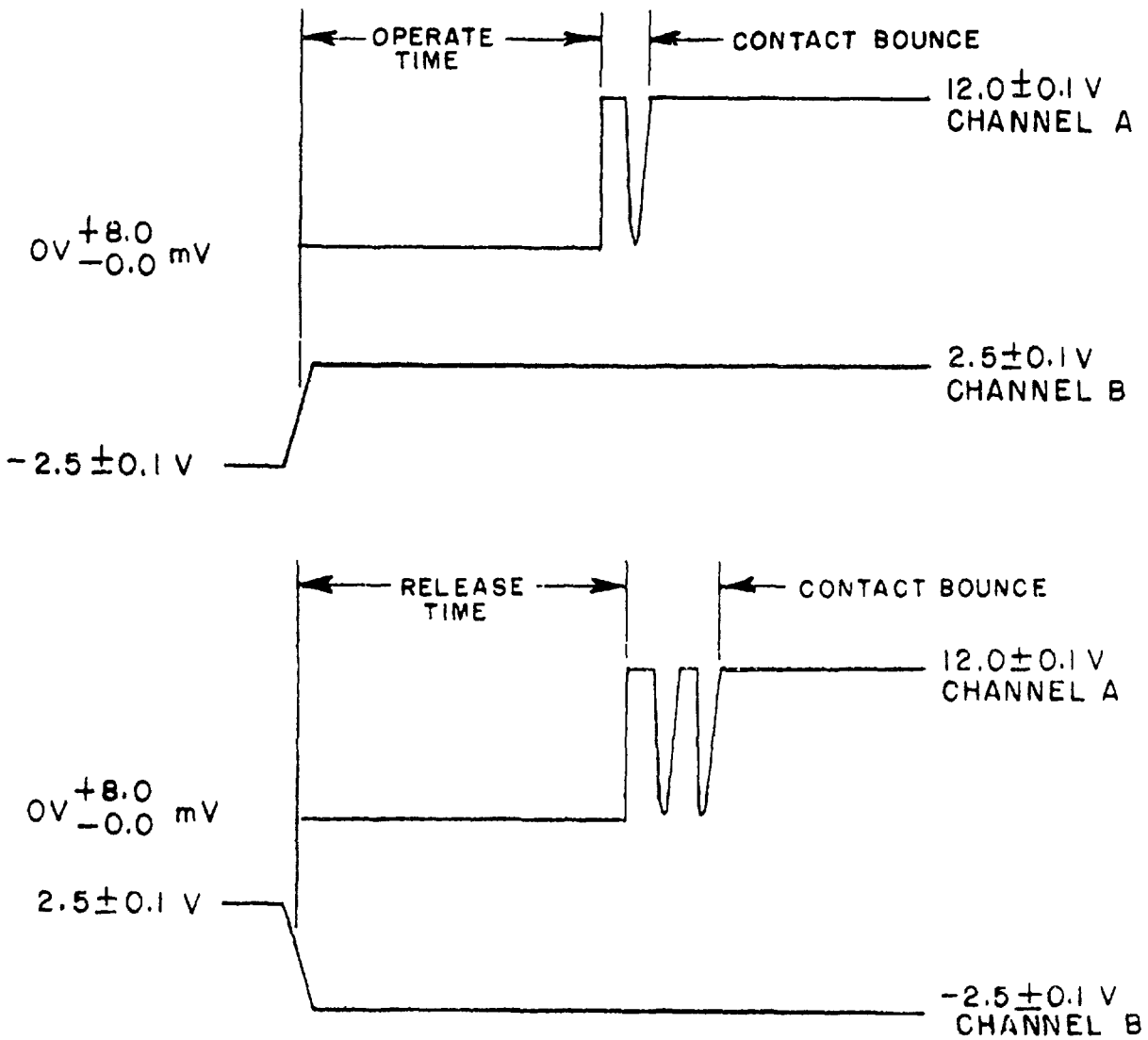
   a. Connect test points A and B as specified for test condition J1.
   b. On item D01, set the trigger coupling to dc, the trigger source to external, the trigger level to positive, and the trigger slope to negative
   c. Adjust output of item H01 for a frequency of 9.0 Hz ±10% and an amplitude of 2.5 ±0.1 Vp as read on item D01, channel B.
   d. The release time shall be read on item D01, channel A, and shall be as specified for test condition J1.
   e. Repeat notes a through d for test conditions J2-J10.

6. Test procedure for test conditions K1-K20:

   a. Connect test points A and B as specified for test condition K1.
   b. On item D01, set the trigger coupling to dc, the trigger source to external, and the trigger level and slope to positive for test conditions K1-K10. The trigger slope shall be set to negative for test conditions K11-K20.
   c. Adjust output of item H01 for a frequency of 9.0 Hz ±10% and an amplitude of 2.5 ±0.1 Vp as read on item D01, channel B.
   d. The contact bounce shall be read on item D01, channel A and shall be as specified for test condition K1.
   e. Repeat notes a through d for test conditions K2-K20.

FIGURE 10. Operate time, release time, and contact bounce test circuit for tables IX, X, and XI - Continued.

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* FIGURE 11. Operate time, release time, and contact bounce test circuit waveforms for tables IX, X, and XI.

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